

# Experiment A: Dynamic vs Static Geometry

The Running Estimator Is Stable; the Static Snapshot Is Chaotic  
3D Is a Property of Firing Events, Not of the Resting Graph  
V20.5 ·  $N = 500$  · 800 sweeps · seed=42

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## Abstract

We measure two  $d_s$  estimators simultaneously on CRF Hybrid V20.5 ( $N = 500$ , 800 sweeps, seed=42):

**Running estimator** — random walks on the KL-graph at the moment of Phase II firing. Biased toward crystallised nodes.

**Static snapshot** — random walks on the full KL-graph at fixed 50-sweep intervals. Includes all nodes, no firing bias.

**Result:**

| Estimator         | Mean  | $\sigma$ | Range      |
|-------------------|-------|----------|------------|
| Running (dynamic) | 3.268 | 0.707    | 2.15–3.95  |
| Static snapshot   | 2.881 | 0.858    | 1.16–3.72  |
| Gap (run–static)  | 0.386 | 1.303    | oscillates |

The key finding is not merely that running  $>$  static. It is that **the running estimator has a stable attractor at  $d_s \approx 3$ , while the static snapshot has no attractor**: it oscillates chaotically between 1.16 and 3.72 with  $\sigma_{\text{static}} = 0.858 > \sigma_{\text{run}} = 0.707$ .

**Conclusion:** 3D geometry is a property of the dynamic KL-structure at firing events. The resting graph carries no stable 3D structure. This confirms the CRF prediction from CRF\_WhereIs3D: geometry is event-defined, not substrate-stored.

**Open:** The gap slope is  $-0.013/\text{sweep}$  ( $R^2 = 0.70$ ), suggesting the static snapshot may converge toward the running estimator at very long runs. This connects to the K35 prediction that  $\xi = \sqrt{\varepsilon_0} = 0.087$  is too small for global propagation (Experiment A-long: 3000+ sweeps needed).

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# 1 Background and Prediction

## Prior Result: CRF\_WhereIs3D

Running V20.5 with isotropic 8D initialisation vs 3D-biased:

- Running estimator (ISO 8D):  $d_s = 3.06$  — consistent with 3D-biased (2.99). Gap: +0.07.
- Static KL snapshot (ISO 8D):  $d_s = 2.57$  — below target. Gap from 3D-biased (3.49): -0.91.

**Interpretation:** 3D is located in the dynamic KL-structure at firing, not in the static graph. The X-positions carry no intrinsic 3D structure.

Experiment A extends this with a time series: does the gap persist over 800 sweeps, or does the static snapshot converge to  $d_s = 3$ ?

**CRF prediction (from K35):** Interaction range  $\xi = \sqrt{\varepsilon_0} = 0.087$ . Mean  $|\Omega_i - \Omega_j|_{\text{mean}} \approx 0.94 \gg \xi$ . Most node pairs are decoupled ( $W_{ij} \approx 0$ ). The Clifford charge transport  $\partial_t C = D\nabla^2 C + S^C$  cannot propagate the 3D structure globally. **Prediction:** static snapshot does not converge to  $d_s = 3$  in finite runs.

# 2 Setup

## Experiment A Setup

| Parameter         | Value                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Simulator         | CRF Hybrid V20.5 ( $N = 500$ , dim=8, $k = 20$ )                                                                            |
| Sweeps            | 800 (burn-in: first 200 sweeps excluded)                                                                                    |
| Seed              | 42                                                                                                                          |
| Running estimator | $d_s$ from <code>_estimate_ds()</code> at each sweep; walks on KL-graph at firing moment; biased toward Phase II nodes      |
| Static snapshot   | Full KL-graph rebuilt from $P$ at sweeps 200, 250, 300, ..., 600; $n_{\text{trials}} = 6$ , all $N$ nodes sampled uniformly |

**Why these are different:** The running estimator walks on the KL-graph *while nodes are firing*, so high- $\Omega$  (Phase II) nodes dominate. The static snapshot walks on the full KL-graph at a fixed moment, averaging over Phase I and Phase II equally.

### 3 Results

#### 3.1 The main finding: stability vs chaos

##### Key Result: Running Stable, Static Chaotic

| Estimator                   | Mean                                           | $\sigma$ | Min  | Max  |
|-----------------------------|------------------------------------------------|----------|------|------|
| Running (dynamic)           | 3.268                                          | 0.707    | 2.15 | 3.95 |
| Static snapshot             | 2.881                                          | 0.858    | 1.16 | 3.72 |
| $\sigma$ ratio (static/run) | 1.21 $\times$ — static is noisier than running |          |      |      |

Both estimators walk on the same KL-graph. The difference is *when*: at firing (running) vs at rest (static).

The running estimator has a clear attractor near  $d_s = 3$ . The static snapshot oscillates between 1.16 and 3.72 with no stable attractor.

**This is the core result:** the 3D attractor exists in the dynamic firing process, not in the graph topology at rest.

#### 3.2 Raw data

| Sweep | Running | Static | Gap    |
|-------|---------|--------|--------|
| 200   | 2.840   | 0.784  | +2.056 |
| 250   | 3.651   | 1.743  | +1.908 |
| 300   | 3.086   | 4.734  | −1.648 |
| 350   | 3.909   | 1.162  | +2.747 |
| 400   | 3.847   | 3.201  | +0.645 |
| 450   | 3.948   | 3.638  | +0.310 |
| 500   | 2.527   | 2.960  | −0.432 |
| 550   | 3.222   | 2.608  | +0.614 |
| 600   | 2.153   | 3.719  | −1.567 |

The static snapshot at sweep 300 reaches  $d_s = 4.734$  — above the theoretical upper bound for 3D geometry. This is consistent with the topology mismatch identified in CRF\_WhereIs3D: a static snapshot during Phase I-dominated periods reflects the high-dimensional background, not the emergent 3D structure.

#### 3.3 Gap trend

Gap mean (sweeps 350–600): +0.386. Gap slope:  $-0.013/\text{sweep}$ ,  $R^2 = 0.70$ .

The negative slope indicates the gap is closing over this window. Two interpretations:

1. **Convergence (K35 prediction fails):** the static snapshot is slowly converging to  $d_s = 3$  via Clifford charge diffusion. Requires long-run test ( $> 3000$  sweeps).
2. **Noise fluctuation:** the gap slope reflects the small number of static measurements ( $n = 6$  post-350) and the high variance of the static estimator. With  $\sigma_{\text{gap}} = 1.303$ , the slope is not yet statistically robust.

The K35 prediction ( $\xi = 0.087 \ll 0.94$ , no global propagation) favours interpretation (2). Resolution requires the long-run test.

## 4 CRF Interpretation

### 3D Is Event-Defined, Not Substrate-Stored

The running estimator measures  $d_s$  at the moment Phase II nodes fire — exactly when the  $\text{SO}(3)$  attractor is active and the KL-field has crossed threshold  $K^*$ .

The static snapshot measures  $d_s$  between events — when the  $\text{SO}(3)$  attractor is not operating and the graph reflects the mixture of Phase I and Phase II nodes.

**The 3D structure is not stored in the graph.** It is re-created at each firing event by the  $\text{SO}(3) \times \Phi$  dynamics. Between events, the graph has no preferred dimensionality.

This is the CRF version of a deep statement: *space does not exist between observations. It is instantiated by the act of becoming real.*

The  $\sigma_{\text{static}}/\sigma_{\text{run}} = 1.21$  ratio quantifies this: the firing process imposes 21% less variance on the dimension estimate than the resting graph. This 21% is the “attractor premium” — the reduction in geometric uncertainty contributed by the  $\text{SO}(3)$  selection mechanism.

## 5 Status Table

| Result                                         | Status     | Note                           |
|------------------------------------------------|------------|--------------------------------|
| Running $d_s$ stable near 3                    | Confirmed  | $\mu = 3.27$ , range 2.15–3.95 |
| Static $d_s$ chaotic (no attractor)            | Confirmed  | $\mu = 2.88$ , range 1.16–3.72 |
| $\sigma_{\text{static}} > \sigma_{\text{run}}$ | Confirmed  | $0.858 > 0.707$                |
| 3D geometry event-defined                      | Confirmed  | runs > static by +0.386        |
| Static can exceed $d_s=4$                      | Confirmed  | sw=300: $d_s = 4.734$          |
| Gap closing (slope $-0.013/\text{sw}$ )        | Hypothesis | only $n = 6$ pts, $R^2 = 0.70$ |
| K35: static does not converge                  | Hypothesis | $\xi = 0.087 \ll 0.94$         |
| Long-run static convergence                    | Open       | 3000+ sweeps, multi-seed       |
| Attractor premium (21%) multi-seed             | Open       | seed=42 only                   |

## 6 Open Questions

### Next Targets

1. **Long-run static convergence (Priority 1).** Run 3000+ sweeps, measure static snapshot every 200 sweeps. If static converges to 3: Clifford charge diffusion overcomes  $\xi$  limitation. If static stays chaotic: K35 prediction confirmed.
2. **Multi-seed reproducibility (Priority 1).** The attractor premium (21% variance reduction) needs 3 seeds to confirm. Single-seed result is suggestive but not conclusive.
3. **Experiment B: SO(3) attractor test (Priority 2).** Inject anisotropy into initial  $\Phi$  phases (CRF-compatible perturbation). Measure recovery of  $d_s = 3$ . If recovered: SO(3) is a true attractor not an artefact of initialisation.

*The resting graph has no preferred dimension.  
Between events, the geometry is unwritten.  
At the moment of firing — when  $D_{KL}$  crosses  $K^*$   
and the SO(3) attractor selects  $n_{\text{lock}} = 3$  —  
three dimensions are instantiated.*

*Space is not a container that persists between observations.  
It is re-created, sweep by sweep,  
in the act of becoming real.*